SSHVisual: Visual Tool to Analyze SSH Data

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***Abstract*—Network operators use visualizations to detect attacks on their systems, interacting with data to uncover patterns that might stay hidden. Operators use these findings to block attackers and to expand the knowledge base of known threats. Detecting anomalies and identifying relationships iteratively in data is, at best, tedious and, at worst, infeasible without data visualizations. Prior work in network visualization analyzed the IP address and ASN dimensions of SSH logins with little focus on the username aspect of the attackers' attempts and detected anomalous network events when baseline events were available. As a result, little attention has been paid to whether attackers that differ in timing, source address, and username belong to the same coordinated campaign. We present SSHVisual, an iterative visualization system evaluated on login attempts collected from our production SSH systems. SSHVisual analyzes the IP, ASN, and username dimensions of SSH attacks, to support campaign detection. It complements automated detection by allowing operators to find anomalous events without a normal baseline. It uses a similarity search across dimensions that groups attackers by behavior and explains each match. It supports iterative and collaborative sharing of findings between operators. We evaluated SSHVisual on 206.6 million attacker login attempts collected over 69 days. We found that it helps operators surface hidden patterns and find attackers to block. In one case study, some IPs tried a similar set of usernames, including uncommon ones, which pointed to a single obscure campaign that volume charts could present as unrelated bursts. In another, similarity search surfaced low-volume IPs working through various username lists, none of which ranked in the top ten by volume. Without similarity search, these IPs would likely have been hard to find and block. These results suggest that username-centric, operator-driven exploration can be valuable because it can reveal covertness and coordination that previous analyses can miss.**

***Index Terms*—SSH, brute-force attacks, attack campaigns, network security visualization, visual analytics, similarity search**

I. Introduction

Computer systems employ SSH [1] to allow users to securely connect to them in order to execute commands and do remote administration. Many users do a username, password combination (password based authentication) in order to gain entry to the SSH system. Attackers, who are adversarial, can use brute force attack mechanisms [2] to get into SSH systems. Sometimes, they guess the username, password combinations and other times, they take advantage of poorly secured systems to build botnets [2] to carry out further attacks [3], [4] or steal resources from the systems, like cryptomining [5], [6].

Given the prevalence of SSH attacks in the IPv4 space [2], we designed SSHVisual and evaluated it on login attempts from our production SSH systems. SSHVisual serves two kinds of network operators: Network administrators, who handle daily SSH traffic, need to identify attackers quickly and block them and Security analysts, who study traffic over longer periods, look for patterns that are not apparent and share what they find. Consider a single period: an administrator can isolate the highest-volume attacker and block it, while an analyst examining the same window can iteratively surface a set of low-frequency attackers that attempt an identical sequence of usernames, and group them as a coordinated campaign.

Surveys of network visualization [7]–[11] establish the maturity of the field and the gaps we target. Our two contributions follow from treating the username, alongside the source IP and ASN, as part of an attacker's identity, and each one serves one of the two operators described above. First, for security analysts, we show that username-based similarity identifies coordinated campaigns that volume-based views split apart. IPs that try the same list of usernames can be recognized as a single campaign, even when they come from different countries and ASNs and are never active on the same day. Second, for network administrators, we show that behavioral similarity search finds attackers that would otherwise stay hidden. Starting from a single IP the administrator already intends to block, it surfaces low-volume IPs that act alike but never reach the top-ten volume rankings, and it explains each match, so the administrator knows which other IPs to block and why. Since network traffic may not offer a stable, attack-free baseline of normal behaviour [12], detectors that rely on such a baseline may miss activity that doesn't deviate from it in the expected way. Operators, by contrast, may recognize coordination and similarity across sessions visually, without specifying normality in advance. The above contributions build on this ability and are meant to complement baseline-driven detection. As operators explore, they can label and share findings so colleagues do not investigate the same patterns twice, and the findings can in turn inform automated detection.

The rest of the paper is as follows: Section II is Related Work, Section III is Tasks and Requirements, Section IV is Design, Section V is Evaluation and Section VI is Conclusion.

II. Related Work

A long line of tools [13]–[20] render network traffic and security events navigable for operators and analysts with the majority being network flow and traffic oriented. Additionally,

a smaller alert- and log-oriented group [21]–[30] organizes IDS output and heterogeneous logs of which [23] contributes a visual analysis (of the CICIDS2017 dataset) that isolates attack clusters—including SSH brute-force (SSH-Patator)—through clustering and dimensionality reduction, namely a multi-resolution wavelet-feature analysis.

A. Attack Campaigns with Time

Honeypot literature does not promote the username identity as an actor in its own right (for attack campaign detection purposes): it remains an attribute to count, never a unit to track—the unit that, alongside the IP and AS, anchors our analysis. Some papers [31]–[37] collect and chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted—the SCADA SSH honeypot [34], kippo-graph [35], and the container deployment [36]—usernames and passwords appear only as static top-N tallies (kippo-graph's top-10 username and password bars are the archetype), never placed on a time axis even where the same tool charts attack volume over time. The WAN honeypot framework [38] is the closest prior system on the credential axis: it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist—a manual, one-off prototype of credential-based coordination analysis, performed on the IP unit and without a temporal dimension.

B. Attacker Similarity with Time

Computed similarity between entities is well established in literature [39]–[42]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on and how it moves through time. On the entity: these systems cluster internal hosts [39], behavioral profiles [41], IP/port members [42], or internal enterprise users grouped by shared resources—not external attacker identities [40]. None treats the username as an attacker actor. On time: [41], [42] reach into the temporal dimension, but both do not compute the similarity of the attacker population across consecutive windows. We apply similarity based metrics—for example, Jaccard overlap of attackers between successive windows—particularly to the username actor unit, capturing how the attacker population persists or churns over time.

C. Limits of Automated Detection

One premise of our approach is that SSH honeypot traffic does not admit the normal baseline that automated anomaly and campaign detection presuppose. The broader literature already documents the fragility of such detection: Sommer and Paxson [43] note that machine-learning anomaly detection is rarely deployed for intrusion detection given the variability of "normal," the cost of errors, and the difficulty of sound evaluation, while Axelsson [44] shows it is throttled by the base-rate fallacy, where the false-positive rate becomes the limiting factor once intrusions are rare relative to benign traffic. Our setting departs from that one entirely, in the direction Gates and Taylor [12] anticipate when they question whether attacks are rare or a stable normal exists, note that normal traffic may itself be malicious and malicious traffic unremarkable, and observe that detection ultimately devolves to an operator investigating clusters of activity by hand. The consequence is visible in the systems cited above: Situ [39] learns a probabilistic model of each IP's behavior and CLIQUE [45] scores each actor against a baseline of its own expected activity, so both presume a per-actor "normal" that near-uniformly hostile, constantly churning traffic cannot supply. Situ nonetheless concedes that automation alone is insufficient—operators "do not want another automated tool with algorithms they do not trust"—and builds a human-in-the-loop system; we reach the same conclusion from a starker starting point, lacking even the per-actor baseline it assumes, and so treat campaign identification as operator-driven from the outset.

D. Username as Actor Identity

We analyze attackers at the level of a campaign, meaning a group of attackers acting together, rather than one attacker at a time. That raises a practical problem: a campaign rarely shows up as a single source address. IP addresses do not stay tied to one machine. One host is assigned many addresses over time [46], hosts cycle through new addresses [47], and counting by address can overstate the number of real machines by orders of magnitude [48]. A view organized by address is therefore likely to scatter one campaign across dozens of entries that look unrelated. The username pulls those pieces back together. Attackers in the same campaign likely draw from a shared credential list, so the same login names recur across the changing addresses. Grouping by username reassembles the campaign that an address-by-address view splits apart.

A username is not a perfect identity either. Names like root are tried by almost everyone, and one campaign runs through a whole list of names. We therefore treat the username as one analysis dimension alongside the address and the network it belongs to (its AS), not as a replacement for them. Uncommon usernames, which unrelated attackers are unlikely to share by chance, serve as the stronger evidence of coordination, and we resolve the remaining overlaps with the similarity and grouping methods described above.

E. Operator Labeling and Collaboration

Since detection on our data is operator-driven, the findings an operator produces must be captured and shared rather than discarded. In two works [19], [49], interactive labeling was used to fold the operator's judgments back into the system. Collaborative analysis has been supported through shared artifacts with [17] and [20] being notable examples. These systems establish that labeling and collaboration are valuable, which is why our labeling-and-sharing mechanism is the step that turns an identified set of usernames into a durable, shared campaign.

III. Tasks and Requirements

The design of SSHVisual draws on the authors' experience as network operators and on gaps we identified in prior work

on SSH attack analysis and security visualization. From these we derive three tasks and five requirements.

A. Tasks

T1: Identify SSH campaigns. Security analysts need to find groups of attackers that behave alike across usernames, IP addresses, autonomous systems (ASes), and countries, and to record those groups so they can be referenced later.

T2: Detect anomalies that warrant action. Network administrators need to notice unusual activity, such as a sudden spike or a single source dominating traffic, quickly enough to decide whether to block it.

T3: Share findings across operators. Analysts and administrators need to see what their colleagues have already investigated, so that the same patterns are not identified twice.

B. Requirements

R1: Attacker similarity profiling. The system should be able to cluster similar attackers based primarily along the username dimension, secondarily along IP and AS information and tertiarily along other dimensions. R1 mostly serves T1 as it allows analysts to build a repository of similar attackers.

R2: Sorting and ranking. SSH data are heavy-tailed: a few usernames [50] and IP addresses [51] account for most attempts, and the remaining entities are buried beneath them. The system should therefore list every entity in each dimension, not only the most active ones, and let the operator sort and rank entities by any combination of attributes. R2 supports T1 and T2: administrators can surface anomalous entities that never reach the top of a volume ranking, and analysts can find entities with similar attribute values.

R3: Shared tagging. Operators should be able to tag attackers, record both what a tag means and why it was applied to a particular attacker, and have those tags persist and remain visible to every operator using the system. Each tag application should record who applied it and when, so that later operators can judge whether a finding is still current. R3 primarily supports T3, and it supports T1 because tags accumulate into a repository of related attackers.

R4: Complementary tabular and temporal views. No single visualization serves every need. A table supports precise lookup and ranking but discards temporal structure; a linechart preserves temporal structure, showing spikes and campaigns at a glance, but conveys values less precisely. The system should provide both views and let operators move between them. An administrator may find and act on an anomaly using the table alone, whereas an analyst may use the table to assemble a set of related attackers and then examine their behavior over time in the linecharts. R4 supports T1 and T2.

R5: Interactive exploration of the linecharts. The linecharts should follow Shneiderman's Visual Information-Seeking Mantra: overview first, zoom and filter, then details-on-demand [52]. They should open with an overview of the full time range, let the operator zoom into a time window and filter which entities are displayed, and reveal details about an individual entity on request. R5 supports T1 and T2:

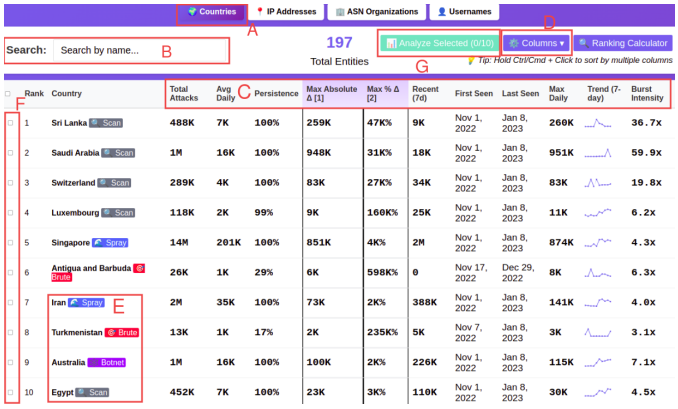


Fig. 1. Discovery table, showing the countries tab (A). Operator can filter any country by name (B). Each row shows a country and operators can sort by any mix of columns (C), with additional columns present in (D). Based on attack patterns, a tag is assigned (E). Operator can select up to ten rows (F and G) to plot in linechart.

zooming isolates the time window of a suspected campaign, and filtering removes dominant entities that would otherwise mask smaller ones. Additionally, line charts become harder to read accurately as more series are added [53]. Each linechart should therefore show at most ten entities.

IV. Design

SSHVisual combines two visual components, a discovery table and a set of linecharts, with two non-visual components, similarity search and shared tagging. We describe the visual design in Section IV-A and the non-visual design in Section IV-B. All figures show SSH login attempts collected between 1 November 2022 and 8 January 2023, a period of 69 days.

A. Visual Design

The interface provides two views of the data. The discovery table (Fig. 1) is time independent: each row summarizes one entity over time. The linecharts (Fig. 2) can be time-dependent. Both views cover the full data period by default.

1) Discovery Table: The discovery table has four tabs, one per dimension: Countries, IP addresses, ASN organizations, and Usernames. In Fig. 1 the Countries tab is active (label A) and the other tabs are inactive. A search box filters rows by name (label B), and the counter above the table reports the number of entities in the current tab (197 countries). Some columns (label C) are shared across all tabs: total attacks, average daily attacks, persistence (the share of days on which the entity was active), the largest absolute and percentage day-over-day change, attacks in the most recent seven days, first- and last-seen dates, maximum daily count, trend (mini line chart showing attacks at 7-day intervals) and burst intensity (ratio of max daily attacks to average daily attacks where a higher ratio likely indicates an entity that attacks sporadically but with higher frequencies).

Some columns (hidden in the dropbox of label D) apply only to particular tabs; Table I in Appendix A lists every

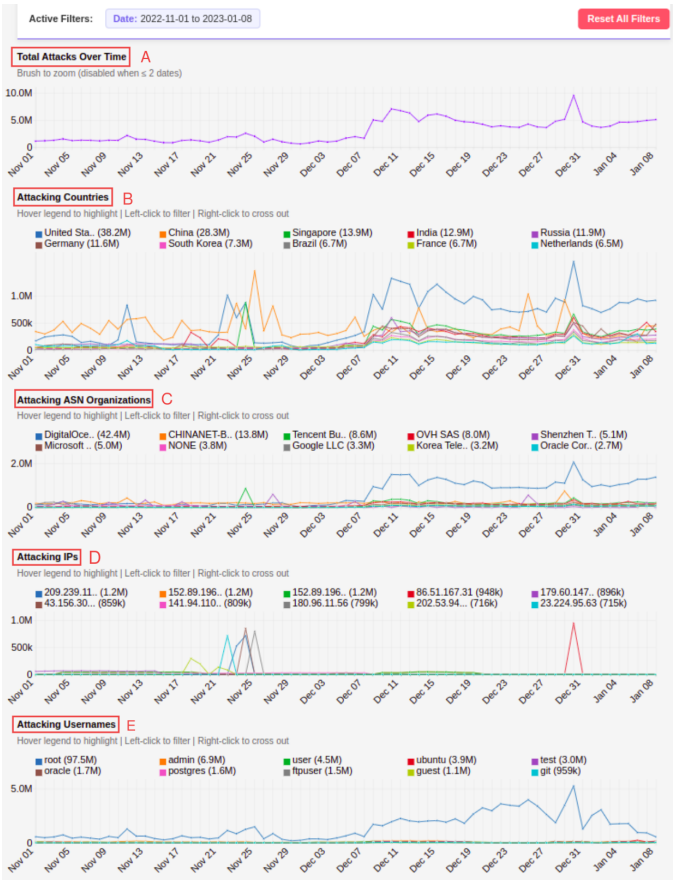


Fig. 2. The aggregate chart (A) at the top shows total attacks over time; the four charts below show the ten most active Countries (B), ASNs (C), IPs (D), and Usernames (E).

column and the tabs it appears in. The IP tab has columns ASN Name and Country that the other tabs lack with the underlying assumption that an IP address belongs exclusively to one ASN and one country. The tabs Countries, ASNs and IPs share these columns: unique usernames, username stability (mean Jaccard similarity between consecutive days' username sets where 0 = volatile and 1 = stable with a higher number indicating that same usernames are used consistently) and username concentration (top 3 usernames and their attack percentages). The tabs Countries, ASNs and Usernames share these columns: unique IPs, IP stability (mean Jaccard similarity between consecutive days' IP sets where 0 = volatile and 1 = stable with a higher number indicating that same IPs are used consistently) and IP concentration (top 3 IPs and their attack percentages). The tabs Countries and Usernames share these columns: unique ASNs, ASN stability (mean Jaccard similarity between consecutive days' ASN sets where 0 = volatile and 1 = stable with a higher number indicating that same ASNs are used consistently) and ASN concentration (top 3 ASNs and their attack percentages). The tabs ASNs and Usernames share these columns: unique countries, country stability (mean Jaccard similarity between consecutive days' country sets where 0 = volatile and 1 = stable with a higher number indicating that same countries are attacking consistently) and country concentration (top 3 countries and their attack percentages). Finally, the ASN tab has a unique column called primary country which shows the country it mostly comes from.

Exclusive to the countries tab, each country entity also carries a badge summarizing its dominant behavior, namely Scan, Spray, Botnet or Brute (label E). Brute marks entities whose traffic comes from few IP addresses making many attempts on few usernames. Spray marks traffic from many IP addresses trying many usernames, and Botnet marks traffic spread across many IP addresses and many ASNs. Scan marks a general attack pattern that fits none of the other three.

Unlike the linecharts, which show only the ten most active entities per dimension, the table lists every entity. Operators can select up to ten rows with the checkboxes (label F) and use Analyze Selected (label G) to plot those entities in the linecharts, and they can return to the linecharts at any time. This link lets operators move from a precise, ranked lookup to the temporal behavior of the same entities, fulfilling R4.

2) Linecharts: The linechart view (Fig. 2) stacks five charts on a shared daily time axis. The first, Total Attacks Over Time, shows the aggregate volume of attempts; the remaining four show the ten most active countries, ASNs, IP addresses, and usernames. Each legend entry lists the entity's total for the selected period. Interaction with the linecharts follows Shneiderman's mantra [52] (R5), and the charts are coordinated so that an action in any one of them updates all of them.

Overview first. By default, the charts cover all sixty nine days of data, with each per-entity chart showing the ten most active entities in its dimension.

Zoom. Every chart supports brushing. When the operator brushes a time range on any chart, the range is applied to all five charts and recorded in the Active Filters bar, so the temporal context stays the same whichever dimension the operator is examining. Brushing is disabled once the range is two days or fewer, since a line needs more points to show a trend.

Filter. Right-clicking a legend entry crosses the entity out, removing it from its chart and rescaling the y-axis so that smaller series are no longer flattened against the axis by a dominant one. Left-clicking a legend entry selects that entity and filters every chart to the attacks associated with it. Because each attack attempt links a username, an IP, an AS, and a country, the selection propagates along these relationships:

- Selecting a country restricts the AS, IP, and username charts to the entities observed in attacks from that country
- Selecting an AS shows the country it originates from, along with its IP addresses and the usernames they try
- Selecting an IP address shows its country and AS, along with the usernames it tries
- Selecting a username shows the countries, ASes, and IP addresses from which it was tried

Details-on-demand. Hovering over a data point shows a tooltip with that entity's attack count for the day, hovering over a legend entry highlights the entity's line, and the legend reports each entity's total for the selected period.

Linked filtering turns the linecharts from four separate rankings into a means of tracing an entity across dimensions. An operator investigating the spike on 30 December, for example, can select `root` in the username chart; the IP, AS, and country charts then show which addresses, networks, and countries tried `root`, and brushing around 30 December narrows the view to the spike itself. Conversely, selecting an AS such as DigitalOcean reveals which usernames its hosts try, indicating whether traffic from a hosting provider reflects one campaign or many. Brushing on every chart, linked filtering, and on-demand details together fulfill R5.

The linecharts answer questions that the discovery table cannot, such as "Do the countries in Fig. 1 share the same peaks and valleys?" or "Does one country begin attacking earlier than another?" In Fig. 2, the aggregate chart shows a sharp spike on 30 December, and most of the top ten countries peak on the same day, indicating a coordinated, multi-country event rather than activity from a single source. The charts also reveal differences in timing. China's largest peak falls in late November, weeks before the US' activity rises, and the IP chart shows individual addresses that each spike for a day or two and then fall silent. Neither pattern can be recovered from the table's per-entity totals. Together, Figs. 1 and 2 show how the two views complement each other, fulfilling R4.

3) Sorting and Ranking: Fig. 3 shows the usernames tab sorted on two columns (labels A and B). Operators add sort keys by holding Ctrl/Cmd while clicking column headers, and the bracketed indices in the headers show when the column was clicked.

Sorting matters because the data are heavy-tailed. The username linechart in Fig. 2 is dominated by `root`, which accounts for tens of millions of attempts and flattens every other line. Ranking by day-over-day change instead surfaces usernames (label C in Fig. 3), which are not seen in Fig. 2, several of which are default accounts for specific software or hardware products (`karaf`, for example, is the default account of Apache Karaf). Their totals are far below the top ten, but their activity is concentrated: `karaf` received 77.6K attempts in total, 44.8K of them on a single day, and was active on only 15 of the 69 days. Bursts of product-specific credentials like these suggest campaigns targeting particular devices, yet they are invisible in a volume-ranked chart. Fig. 3 thus shows how sorting and ranking over the full entity list fulfills R2.

B. Grouping and Tagging

The visual components help operators find individual attackers, while the grouping and tagging components help them connect those attackers to others and record what they find. Similarity search groups attackers by behavior, in the username dimension (Section IV-B1) and the IP dimension (Section IV-B2), and shared tagging (Section IV-B3) preserves the resulting groups so that other operators can build on them.

1) Username Similarity: Similarity search lets operators start from one attacker and retrieve the attackers that behave most like it. For usernames, the unit of analysis is the population of sources that tries a given username. For each username we compute the breadth of that population (the number of distinct IP addresses, ASes, and countries that try it), its volume, its persistence, and how concentrated its traffic is on its single most active source. We do not reuse the seventeen-dimensional profile we apply to IP addresses (Section IV-B2), because a diagnostic analysis showed several of those features to be uninformative for usernames. Day-to-day stability is the clearest example: most IP addresses are active on many days, so their stability is well defined, but about half of the usernames appear on only a single day, where stability is meaningless. We cluster the username profiles with k-means, selecting $k = 9$. Each cluster is labeled automatically from its centroid. To answer a query, the system retrieves the query's cluster and ranks the other members by Euclidean distance to the query, reported as a similarity score.

In Fig. 4, an operator searches for the username `james` (label A). The system places it in Cluster 3, labeled High-Volume Persistent Botnet-Scale, which contains 2,624 usernames; the cluster summary reports that a member receives about 31.8K attempts from roughly 4K IP addresses and is active on 72% of days (label B). The results table lists the 20 most similar usernames (5 shown here), each with its total attacks, top country, top three source IPs, the top three ASNs, and a set of "Why similar?" labels (label D), such as a shared lead source country, comparable volume, or sourcing volume, that explain the match.

Above the results, a shared infrastructure panel (label C) lists IP addresses that appear among the top sources of more than one result, and these addresses are shown in the table. In Fig. 4, thirty seven addresses are shared. Each line of the panel names an address and counts how many of the 100 results it is a top source for, followed by examples of

Rank	Username	Total Attacks	Avg Daily	Persistence	Max Absolute Δ [1]	Max % Δ [2]	Recent (7d)	First Seen	Last Seen	Max Daily
1	karaf	77.6K	5.2K	21.70% (15d)	32.4K	360.52K%	120	Nov 2, 2022	Jan 8, 2023	44.8K
2	ossuser	82.6K	2.6K	46.40% (32d)	30.4K	276.15K%	690	Nov 3, 2022	Jan 8, 2023	46.0K
3	snow	248.2K	5.2K	69.60% (48d)	22.0K	168.87K%	41.8K	Nov 1, 2022	Jan 8, 2023	35.2K
4	nmrsu	89.4K	1.8K	72.50% (50d)	35.0K	145.91K%	2.9K	Nov 5, 2022	Jan 8, 2023	43.8K
5	lenovo	25.3K	476.51	76.80% (53d)	8.5K	283.10K%	5.7K	Nov 1, 2022	Jan 8, 2023	8.5K
6	fb	9.7K	225.67	62.30% (43d)	4.7K	470.10K%	403	Nov 5, 2022	Jan 8, 2023	4.7K
7	inspur	40.4K	684.71	85.50% (59d)	5.3K	263.05K%	11.2K	Nov 1, 2022	Jan 8, 2023	5.3K
8	radware	12.3K	686.11	26.10% (18d)	4.8K	241.25K%	1.9K	Nov 1, 2022	Jan 8, 2023	4.8K

Fig. 3. Usernames tab sorted by Max Absolute Δ (A) and Max % Δ (B), surfacing bursty usernames (C).

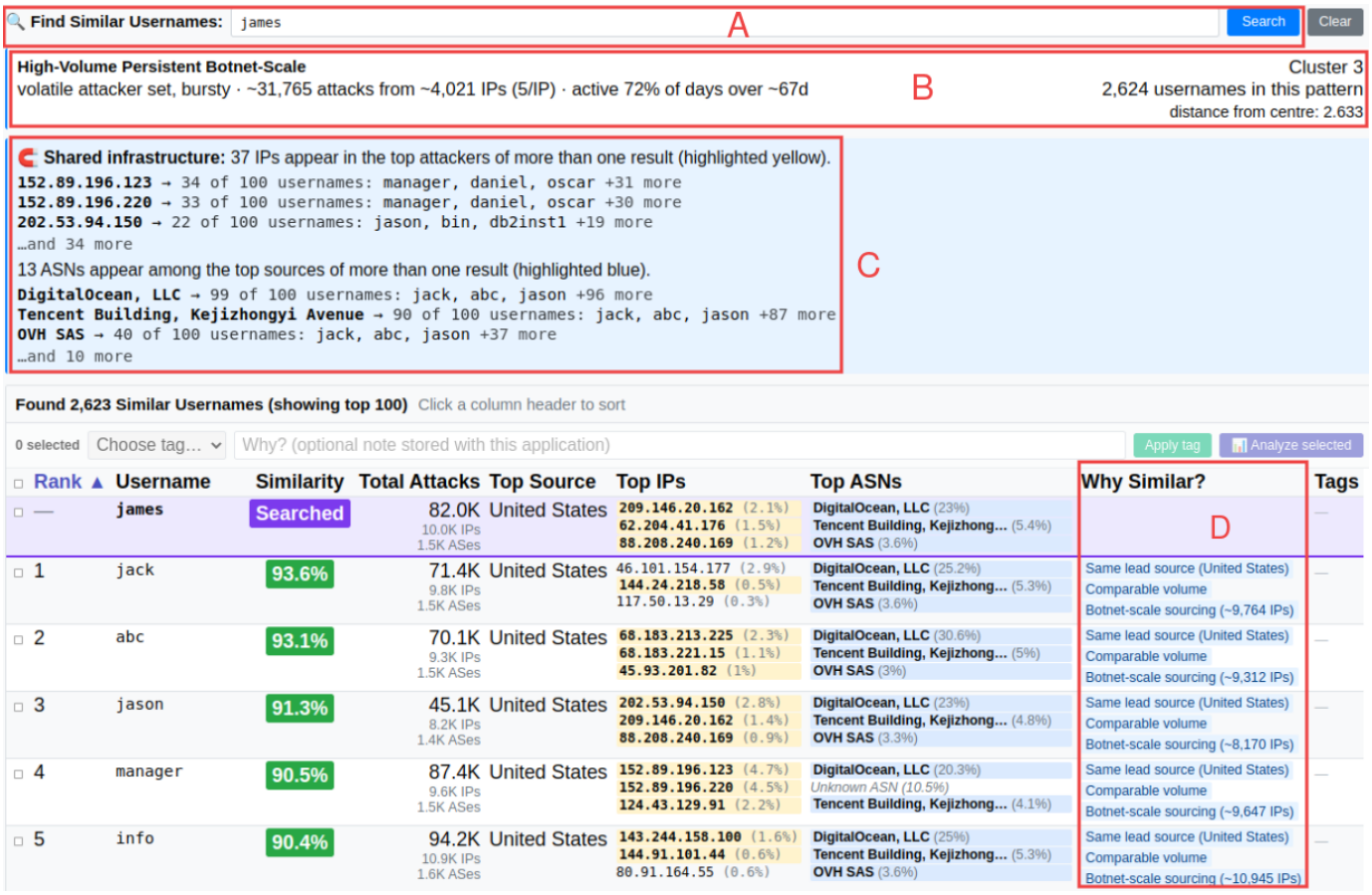


Fig. 4. Similarity search for username `james` (A). The cluster summary (B), shared infrastructure panel (C), and "Why similar?" (D) labels explain why the results were grouped with the query.

those usernames. Two of the most shared addresses (boxed in Fig. 4), 152.89.196.123 and 152.89.196.220, lie in the same /24 network and are top sources for 34 and 33 results respectively, including `manager`, `daniel`, and `oscar`. The panel applies the same analysis to ASes: 13 are shared, led by DigitalOcean (99 of 100 results), Tencent (90), and OVH (40). The panel turns behavioral resemblance into infrastructural evidence: usernames tried by the same hosts are likely drawn from the same credential list and belong to the same campaign.

2) IP Address Similarity: We characterize each IP address by a seventeen-dimensional profile drawn entirely from its activity during the data period. The features fall into four groups: volume (log-scaled totals, daily means, and peaks); temporal shape (the fraction of days on which the address was active, its burst intensity, its largest day-over-day swings, a fitted trend slope, its activity span, and the share of its traffic in the final week); targeting behavior (the number of distinct usernames it tries, how stable that set is from one day to the next, its rotation rate, and the share of attempts taken by its most-tried username); and two flags indicating whether the address belongs to a cloud provider and whether it originates from a major source country. We cluster these profiles with k-means, label each cluster from its centroid, and answer queries in the same way as for usernames.

In Fig. 5, an operator searches for 61.177.173.46 (label A), which falls in Cluster 5, labeled Low-Volume Persistent Exploratory, containing 4,405 addresses. The most similar results are shown in label B, with similarity scores between 91.2% and 80.4%, all belonging to CHINANET-BACKBONE. Behavioral features alone therefore recover infrastructure relationships, suggesting that these hosts are operated together.

Figs. 4 and 5 show how SSHVisual lets operators assemble sets of similar attackers along the username and IP dimensions and see why those attackers were grouped, fulfilling R1.

3) Shared Tagging: Figs. 6–8 show how an operator records a finding so that other operators can see work that has already been done. Tags are organized in a tag repository for each dimension. The repository displays the current operator's name, and every tag and tag application records its author.

In Fig. 6 the operator creates a tag, "similarity to james" (label A), for the username cluster found in Fig. 4. The repository now lists the tag with its description (label B) and its author (label C). Tags are applied from the similarity results. In Fig. 7 the operator selects seven usernames (label A), chooses the new tag, and adds a note explaining this particular application: "recurring ASNs" (label B). In Fig. 8 the repository reports seven applications (labels A and F).

Filtering the list by tag shows each tagged username with its note (labels B and C), the operator who applied it (label D), and a timestamp (label E), and any application can be removed with Untag. Tagged usernames also appear in the Tags column of subsequent similarity searches, so an operator who later retrieves the same usernames sees immediately that they have already been investigated.

Separating a tag's meaning from the rationale for each application lets a later operator understand both what a group represents and why each member belongs to it, without repeating the analysis. The recorded author and timestamp indicate whether a finding is recent. Figs. 6–8 thus show how tagging supports collaboration among operators, fulfilling R3.

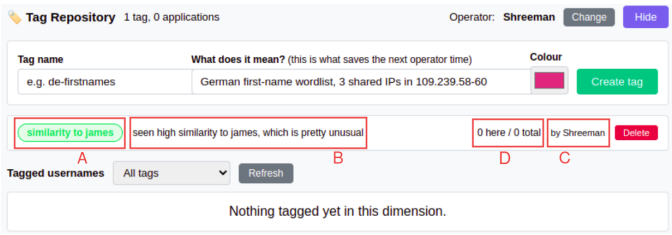


Fig. 6. A newly created tag (A), listed with its description (B), author (C), and application counts (D).

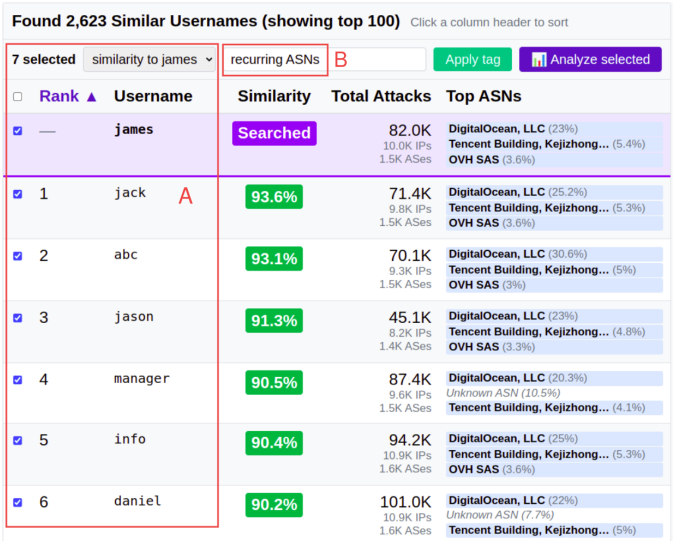


Fig. 7. Applying the tag to seven usernames from the similarity results (A), with a note explaining the application (B).

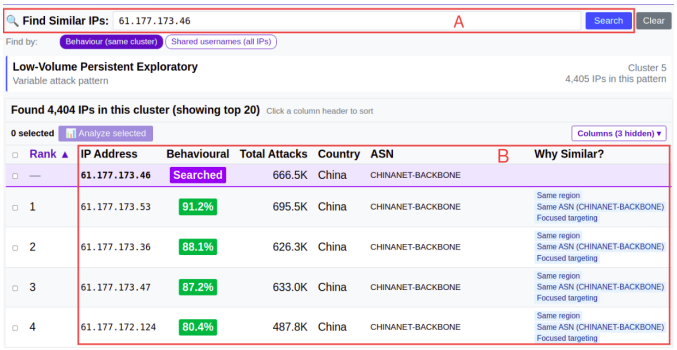


Fig. 5. Similarity search (A) for IP 61.177.173.46. Nearest neighbors are adjacent addresses on the same network (B).

V. Evaluation

We evaluate SSHVisual on its ability to uncover patterns. We show this in two ways: 1) finding campaigns in which the same list of attempted usernames recurs across IPs in different countries and 2) finding entities to block that act alike but would be buried underneath entities that dominate attacks. Both the subsections start with a network operator coming across a single IP, for example from an intrusion detection alert, an abuse report, a flag report on stackoverflow or the discovery tables in SSHVisual. On its own, such an IP is one address among tens of thousands; SSHVisual helps the operator find what else it is connected to.

A. Username Campaigns

Fig. 9 shows the username search for IP 68.183.213.225 (DigitalOcean, Germany). Four IPs score 100% username similarity with it (labels A–E): 143.244.183.240, 24.199.108.32 and 146.190.113.228 in the United States, and 134.122.62.125 in the Netherlands (not shown in Fig. 9 but at least 12 IPs have a username similarity score greater than 84%). All five try the same 77 usernames in near-identical proportions, and all 16 of the searched IP's uncommon usernames (such as ecell, guojing and xxaifwq) appear in every list. The uncommon usernames matter here: a 100% score built on common usernames like

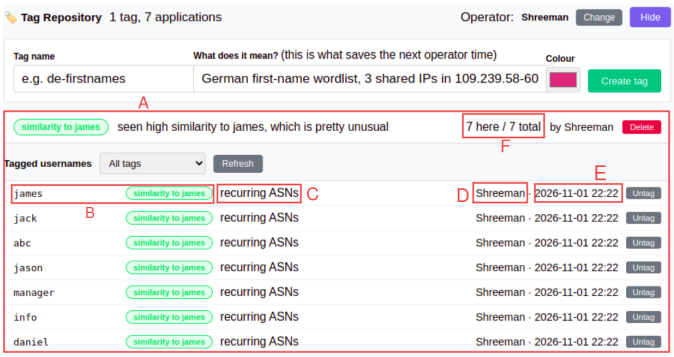


Fig. 8. The tag repository after the tag is applied (A), listing each tagged username (B) with its note (C), author (D), timestamp (E), and application counts (F).

	Rank	IP Address	Usernames	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
	—	68.183.213.225 A	—	its top usernames ecell guojing xxaifwq +74 more	16 its uncommon usernames ecell guojing xxaifwq +13 more	373.6K	Germany	DigitalOcean, LLC
	1	143.244.183.240 B	100%	ecell guojing xxaifwq +74 more	16 ecell guojing xxaifwq +13 more	136.4K	United States	DigitalOcean, LLC
	2	24.199.108.32 C	100%	guojing escheduler ecell +74 more	16 guojing escheduler ecell +13 more	140.5K	United States	DigitalOcean, LLC
	3	146.190.113.228 D	100%	guojing ecell xxaifwq +74 more	16 guojing ecell xxaifwq +13 more	155.4K	United States	DigitalOcean, LLC
	4	134.122.62.125 E	100%	ecell escheduler guojing +74 more	16 ecell escheduler guojing +13 more	159.1K	Netherlands	DigitalOcean, LLC

Fig. 9. 100% usernames shared among five IPs. IPs A, B, C, D, E match the IPs A, B, C, D, E in Fig. 10.

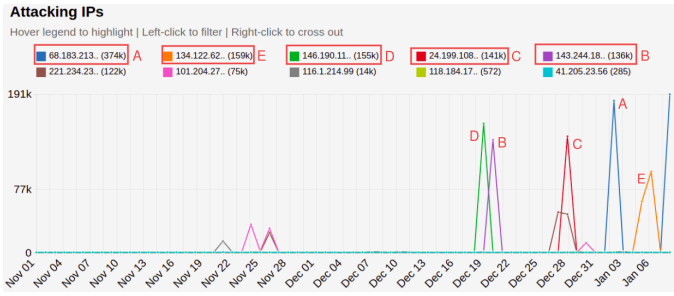


Fig. 10. Five IPs labeled A–E from Fig. 9 in the linechart, along with other IPs with almost the same username list.

root and admin would mean little, but 16 shared uncommon usernames are unlikely to occur by chance. The five IPs sit on one ASN across three countries (not shown in Fig. 9 but the 12 IPs mentioned before come from 5 countries and 6 ASNs), which is consistent with a single operator renting servers in several regions.

Fig. 10 shows why this campaign is hard to see from volume alone. Each of the IPs is active for only a day or two, and at different times: 146.190.113.228 on Dec 19 (label D), 143.244.183.240 on Dec 20 (label B), 24.199.108.32 on Dec 28 (label C), 68.183.213.225 on Jan 2 and Jan 8 (label A), and 134.122.62.125 on Jan 5 to 6 (label E). The lines never rise together, so the chart suggests five unrelated bursts. Only the shared username list links them into one campaign.

	Rank	IP Address	Behavioural	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
	—	200.195.169.59	Searched	its top usernames oix jqz alu +3818 more	925 its uncommon usernames oix jqz alu +922 more	36.4K	Brazil	Ligga Telecomunicações S.A.
	1	132.255.253.123	94.9%	nw noroot elloise +1799 more	46 elloise alameda tab +43 more	34.8K	Brazil	BRASIL TECPAR SERVICOS DE TELECOMUNICACOES SA
	2	103.186.99.250	92%	final maison virginie +1681 more	72 final maison virginie +69 more	29.5K	Indonesia	PT Network Center Nusantara
	3	79.3.19.34	90.8%	sdt mallika yfc +1896 more	83 sdt helix cch +80 more	36.2K	Italy	Telecom Italia S.p.A.
	4	201.116.12.217	89.2%	nash mule vojta +1863 more	140 nash vojta teamspeak-server +137 more	33.8K	Mexico	UNINET
	5	27.99.9.141	88.6%	anusha andri prabhu +1839 more	86 damares milo libevent +83 more	41.2K	Australia	Microplex PTY LTD

Fig. 11. Five IPs that behave like 200.195.169.59.

B. Finding Hidden Entities to Block

Fig. 11 shows the IP similarity search for 200.195.169.59, an IP from Ligga Telecomunicações S.A. in Brazil with 36.4K attacks. The five closest matches (88.6% to 94.9% behavioral similarity) come from Brazil, Indonesia, Italy, Mexico and Australia, each from a different ASN (not shown in Fig. 11 but the top 20 similarity matches—at least 86%—to this IP come from 15 different countries and 21 different ASNs). Despite this spread, they behave alike: each made roughly 30K to 41K attacks, is tagged as exploratory targeting (that is, they use many different usernames), and shares between 1,684 and 1,899 usernames with the searched IP, 46 to 140 of which are uncommon (tried by fewer than 1% of all IPs). The searched IP itself tried 3,821 usernames, 925 of them uncommon. This is the pattern: low-volume IPs working through large and similarly uncommon username lists from many countries at once.

Fig. 12 shows how buried these IPs are despite their interesting nature to administrators. None of the 10 IPs (selected from Fig. 11) would appear in the top volume IPs chart. The searched IP ranks #296 of 86,985 IPs and accounts for 0.018% of 206.6M attacks. Its country, Brazil, is large enough to appear in the top 10 countries (#8), but its ASN is 13 times too small to appear in the top 10 ASNs, and the IP itself is 20 times too small to appear in the top 10 IPs. The other IPs rank between #222 and #2,826. An administrator looking only at the high volume charts would see Brazil but never these IPs. With SSHVisual, administrators know not just the IP they found to block but also similar IPs that they should block, and most importantly, why they should block them.

VI. Conclusion

We presented SSHVisual, a visual analytics system that treats the username as a first-class analytic dimension along-

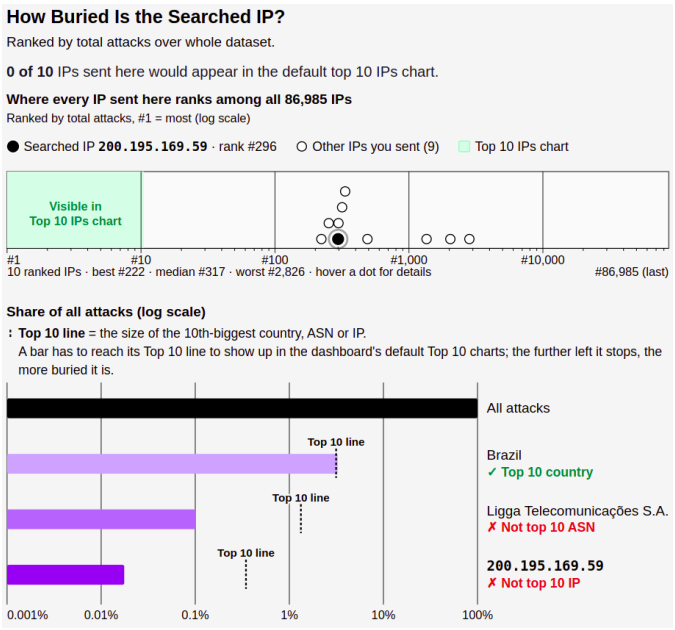


Fig. 12. How buried 200.195.169.59 and its similar IPs are.

side the IP address, AS, and country in SSH attack data. SSHVisual pairs a discovery table that ranks every entity with coordinated linecharts that follow Shneiderman's mantra, so operators can move between precise lookup and temporal behavior. Similarity search groups attackers by behavior along the username and IP dimensions and explains each match. Shared tagging preserves those groups so that findings persist and remain visible across operators. Our case studies show that this combination surfaces patterns that volume-based, IP-centric views miss. Low-volume IPs spread across many countries and ASes worked through similar lists of uncommon usernames, yet none ranked high enough to appear in any top-ten chart. IPs that never attacked on the same day were linked into a single campaign only by the 77 usernames they shared. Because SSH honeypot traffic offers no normal baseline for automated detection, these results suggest that exploratory, operator-driven analysis remains a necessary complement to automated methods, and that the username is a valuable lens for recognizing coordinated attackers.

APPENDIX A
DISCOVERY TABLE COLUMNS

Table I lists every column of the discovery table (Section IV-A1) and the tabs in which it appears.

TABLE I
DISCOVERY TABLE COLUMNS BY TAB

Column	Countries	IPs	ASNs	Usernames
Total attacks	✓	✓	✓	✓
Average daily attacks	✓	✓	✓	✓
Persistence	✓	✓	✓	✓
Max absolute Δ	✓	✓	✓	✓
Max % Δ	✓	✓	✓	✓
Attacks in last 7 days	✓	✓	✓	✓
First / last seen	✓	✓	✓	✓
Max daily count	✓	✓	✓	✓
Trend	✓	✓	✓	✓
Burst intensity	✓	✓	✓	✓
Unique usernames	✓	✓	✓	
Username stability	✓	✓	✓	
Username concentration	✓	✓	✓	
Unique IPs	✓		✓	✓
IP stability	✓		✓	✓
IP concentration	✓		✓	✓
Unique ASNs	✓			✓
ASN stability	✓			✓
ASN concentration	✓			✓
Unique countries			✓	✓
Country stability			✓	✓
Country concentration			✓	✓
ASN name, Country		✓		
Primary country			✓	
Behavior badge	✓			

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